

1. Give an example of a function $f : \mathbb{R} \rightarrow \mathbb{R}$ that is discontinuous at $x = 2023$. Justify.
2. Prove by definition that

$$\lim_{x \rightarrow 1} x^2 + 2x + 2020 = 2023.$$

3. This is a *false* statement.

If $|f(x) - L| < \varepsilon$ for all x for which $0 < |x - a| < \delta$, then $|f(x) - L| < \varepsilon/2$ for all x for which $0 < |x - a| < \delta/2$.

Prove that this is false by finding a function f and numbers a and L which are a counterexample. Make sure to state f 's domain and codomain.